

FINAL YEAR B.TECH PROJECT · 2024-25

# Sign Bridge

Sign to Multi-Speech Converter

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# Introduction

70M+



deaf individuals worldwide rely on sign language as their primary mode of communication.



## Sign Language

A visual-spatial language using gestures and hand shapes as vocabulary.



## Communication Gap

Deaf individuals face daily barriers when interacting with hearing communities.



## Accessibility Need

Automated real-time translation can eliminate these barriers affordably.

# Problem Statement



## No Real-Time Translation

Existing tools lack the ability to translate sign language in real time from a webcam feed.



## Hardware Dependency

Sensor gloves and specialized hardware are expensive, limiting mass adoption.



## Language Isolation

Most systems support only English output, ignoring multilingual needs of India.



## High Latency

Current deep learning pipelines are too slow for smooth, fluid conversations.

# Motivation

*"Language should never be a barrier to human connection."*



## 1 in 6 People

Over 1 billion people globally live with some form of disability. Sign language users are among the most underserved.



## India's 18M Deaf

India has one of the world's largest deaf populations, yet almost no affordable sign language technology.



## Zero Cost Access

A browser-based system removes all hardware barriers — only a webcam is needed.

# Objectives

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## Real-Time Recognition

Capture and classify ISL gestures live via webcam with <200ms latency.



## Multilingual Translation

Convert recognised text to Hindi, Marathi, French, Spanish & more.



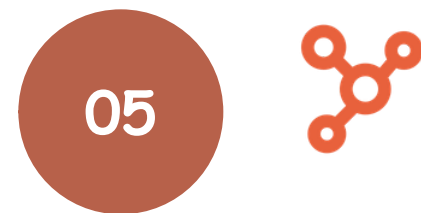
## Speech Synthesis

Generate natural audio playback in the chosen target language.



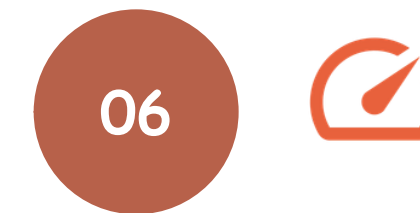
## Privacy-First Design

All processing runs locally on the server — no data leaves the machine.



## Browser-Native Access

Zero installation required; fully functional in any modern browser via WebRTC.



## Low Latency Pipeline

Async FastAPI backend with Uvicorn for maximum throughput.



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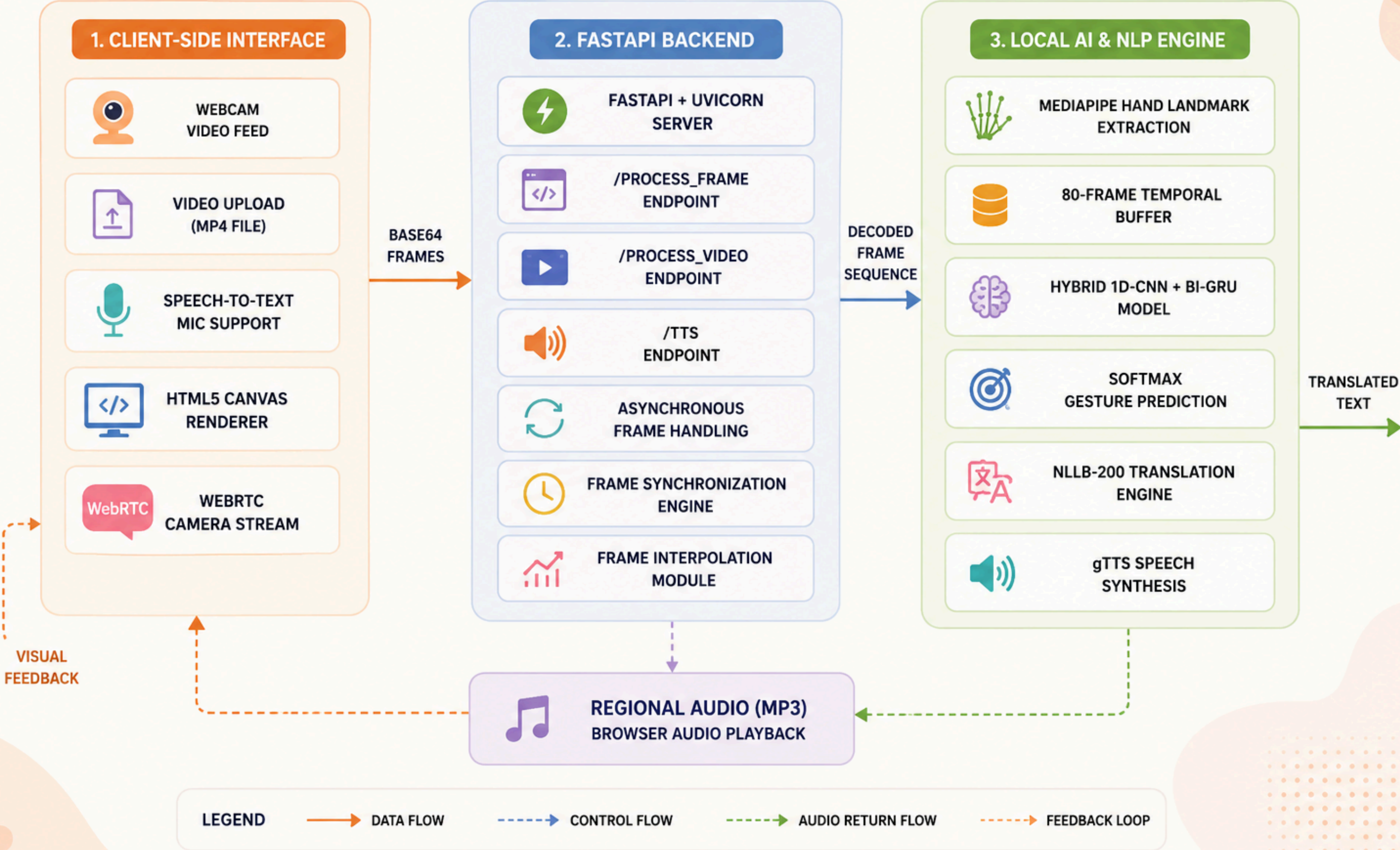
# System Design

Objectives · Existing System · Proposed System · Tech Stack



# SYSTEM DESIGN DIAGRAM – SIGNBRIDGE

Sign to Multi Speech Converter



SECTION

# Dataset and Methodology

MediaPipe · CNN-GRU · Temporal Buffers · Training



# Dataset Description

61

**Gesture Classes**

ISL common words

80

**Frames / Sample**

Temporal window per gesture

21

**Landmarks / Hand**

3D (x, y, z) per keypoint

63

**Features / Frame**

21 × 3 coordinate vector

The Indian Sign Language (ISL) Video Dataset is a video-based gesture dataset containing multiple ISL alphabet and word gestures collected from different contributors. The dataset was used to train the model by extracting skeletal hand landmarks using MediaPipe and generating temporal gesture sequences for real-time sign language recognition.

Each sample consists of 80 temporal frames with 42 hand landmarks (126 coordinate features) extracted using MediaPipe and fed into the Hybrid 1D-CNN + Bi-GRU model.

# MediaPipe Landmark Extraction

## Why Landmarks Instead of Raw Images?



### Lighting Invariant

Pure geometry — unaffected by skin tone, lighting or background clutter.



### Ultra-Lightweight

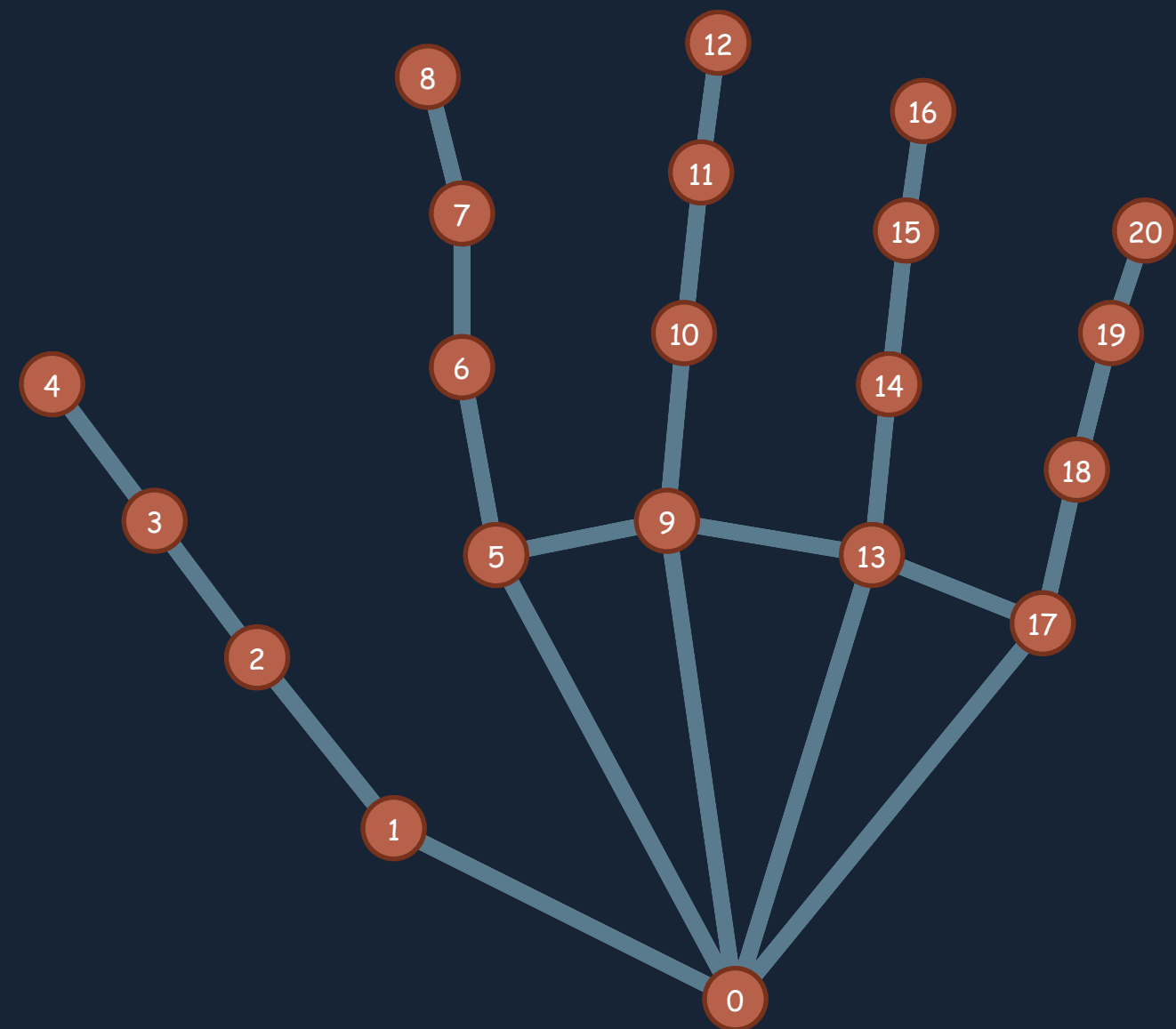
63 floats per frame instead of 307,200 raw pixels — 5,000× smaller.



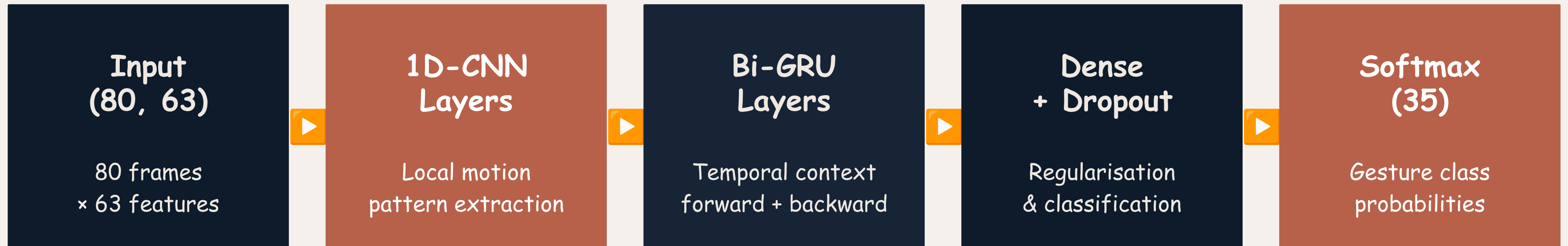
### 3D Depth Encoding

MediaPipe returns x, y, z enabling depth-sensitive gesture disambiguation.

### 21 Hand Landmarks



# Hybrid Deep Learning Architecture



## 1D Convolutional Neural Network

Scans the temporal dimension to extract local motion patterns — joint velocity bursts that characterise distinct gesture segments.



## Bidirectional GRU

Processes the sequence both forward and backward, capturing context that a unidirectional model would miss. Crucial for gestures that loop.



## Why Hybrid?

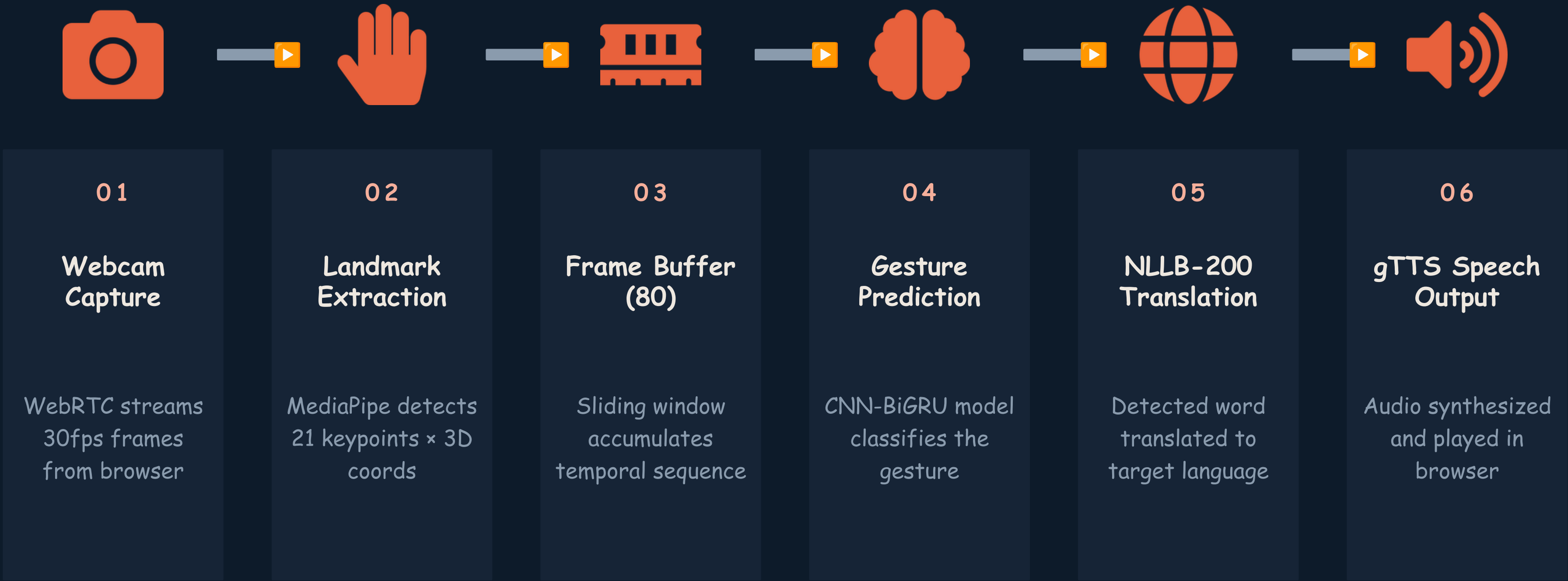
CNN captures local features; GRU models global temporal structure. Together they outperform either alone on gesture sequences with variable execution speed.

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# Backend & Pipeline

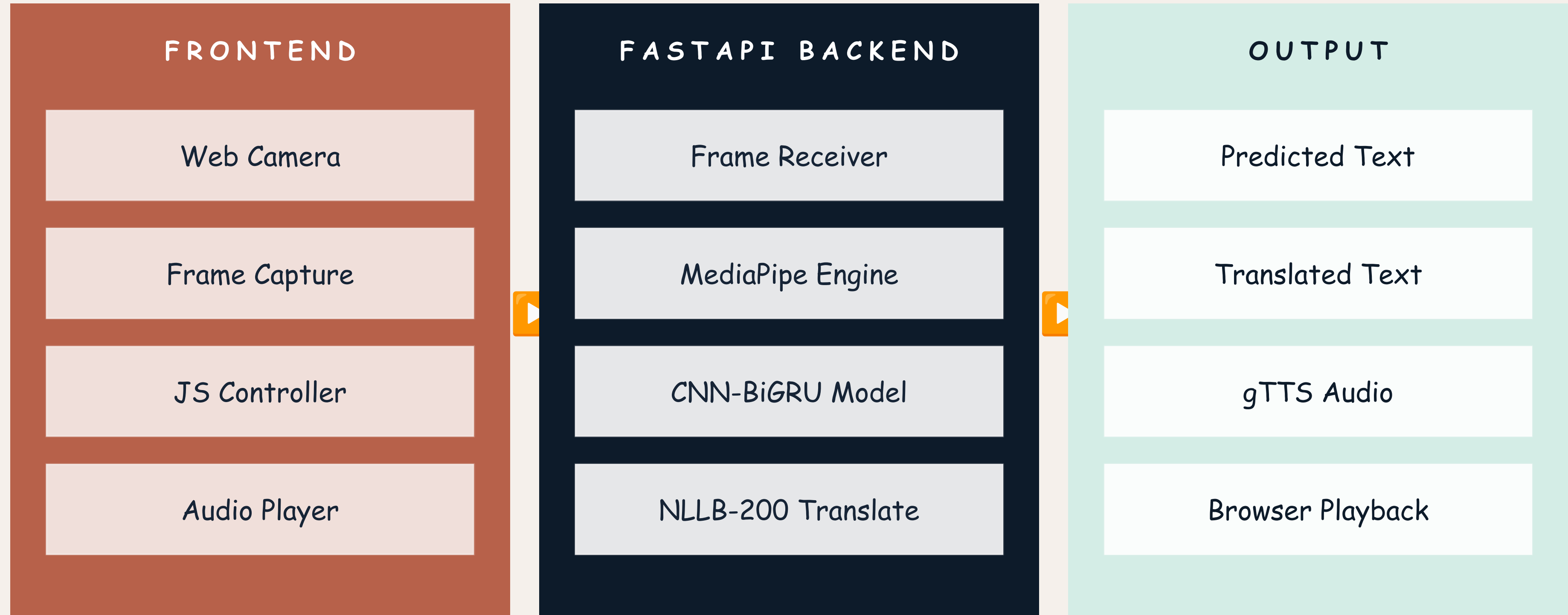
Frame Sync · FastAPI · Translation · Speech

# System Workflow





# System Architecture





# Translation & Speech Synthesis

## Meta NLLB-200

No Language Left Behind — 200-language neural translation model by Meta AI.

Hindi (hin\_Deva)

Marathi (mar\_Deva)

French (fra\_Latn)

Spanish (spa\_Latn)

German (deu\_Latn)

Tamil (tam\_Taml)

Telugu (tel\_Telu)

Gujarati (guj\_Gujr)



## Google Text-to-Speech

- ✓ Python gTTS library wraps Google TTS
- ✓ Generates MP3 in-memory (io.BytesIO)
- ✓ Streamed directly to browser via HTTP
- ✓ HTML5 <audio> auto-plays on receive
- ✓ Supports all NLLB output languages

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# Results

Website Walkthrough · Predictions · Accuracy

# Website Working



## Open Browser

User navigates to localhost or deployed URL. No install needed.



## Activate Webcam

Browser prompts camera permission. WebRTC stream begins at 30fps.



## Perform Gesture

User signs a word. 200 frames captured and normalized.



## AI Prediction

CNN-BiGRU classifies gesture. Confidence score displayed.



## Translation

Predicted word sent to NLLB-200. Target language selected.



## Speech Playback

gTTS generates audio. Browser plays automatically.

# Results & Outputs

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91.2%

**Overall Accuracy**

61-class gesture set

<150ms

**Avg. Latency**

Prediction + translation

9

**Output Languages**

NLLB-200 support

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# Conclusion

*Advantages · Limitations · Future · Summary*



# Advantages & Limitations

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## What Works Well

- ✓ Real-time prediction under 150ms
- ✓ Zero hardware cost — webcam only
- ✓ 9 multilingual output support
- ✓ Privacy-preserving local inference
- ✓ Browser-native, no install required
- ✓ Lightweight landmark-based input

## Known Limitations

- ⚠ 61-gesture vocabulary (no full sentences yet)
- ⚠ Sensitive to poor lighting conditions
- ⚠ Performance varies with camera quality

# Future Scope



F1

## Continuous Sentence Recognition

Transition from isolated word recognition to full natural-language sentence formation using CTC.



F2

## Mobile Deployment

Port the model to TFLite for Android/iOS. Inference on-device for offline, low-latency use anywhere.



F3

## Pose & Face Tracking

Incorporate MediaPipe Holistic for body pose and facial expression — enabling full ISL grammar encoding.



F4

## Larger Gesture Vocabulary

Expand dataset to 500+ gestures covering complete ISL alphabet, numbers, phrases and daily expressions.



F5

## Transformer-Based Model

Replace CNN-BiGRU with a Vision Transformer (ViT) or Temporal Transformer for richer sequence modelling.



F6

## Community Training Platform

Crowdsource gesture recordings from the deaf community to improve model diversity and regional variation.



# Conclusion

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SignBridge successfully demonstrates that AI-powered real-time sign language translation is achievable with consumer hardware and open-source tools.

- 91.2% accuracy on 61 ISL gesture classes

- <150ms end-to-end latency

- 9+ language multilingual output

- 100% webcam-only, no extra hardware

- Open-source, deployable in minutes



# Thank You

Questions & Discussion

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